

Game Makers — Week 3

Loops and Randomness

Name: _____

TODAY'S BIG IDEA

If you're about to do the same thing twenty times — **don't**. Tell the computer to repeat it.

```
repeat 4
  clap
  clap
  stomp
```

That's 12 actions written in 4 lines. Change the 4 to a 100 and you've just done 300 actions by typing two characters.

And when you want the computer to surprise you, use `pick random`.

WORDS TO KNOW

Word	What it means
loop	Code that repeats
repeat n	Do this exactly n times
forever	Do this until the game stops. Nothing can go after it
random	A number the computer picks that you can't predict

⚠ **Golden rule:** every `forever` loop needs a `wait` inside it somewhere. Without one, Scratch freezes and you lose your work.

CLASS ACTIVITY A — DANCING SPRITE

```
when green flag clicked
  forever
    next costume
    wait 0.2 seconds
    move 5 steps
```

Change the `0.2` until the walk looks right. Add `if on edge, bounce` so it doesn't escape.

CLASS ACTIVITY B — RAIN

```
when green flag clicked
  forever
    go to x: (pick random -240 to 240) y: 180
    repeat 20
      change y by -18
      wait 0.02 seconds
```

A raindrop that appears somewhere random along the top and falls. Forever.

Now change the numbers. Fewer repeats + bigger step = faster. Try it.

YOUR PROJECT — THINGS FALL FROM THE SKY!

Today your project starts looking like a real game.

- [] Add a falling object sprite that fits your theme
- [] Make it the right size
- [] Give it the falling script (same shape as Rain)
- [] **Tune the speed** so it's catchable but not boring
- [] **Save it**

It doesn't notice your player yet — you can "catch" it and nothing happens. That's next week.

Homework

1. A second thing to catch. Add another falling sprite that looks different. (*Shortcut: right-click your first one and duplicate it — the code comes with it!*)

2. Unpredictable speed. Use `pick random` so each drop falls at a different speed. (*Hint: the `wait` can take a random number.*)

Bonus challenges

1. Draw shapes with `repeat` + `turn`. A square is `repeat 4 [move 100, turn 90]`. What makes a triangle? A hexagon? What does `repeat 36 [move 10, turn 10]` draw?
 2. Make your falling object spin as it falls.
 3. Make a starfield — five objects falling at five different speeds.
 4. Use `repeat until` instead of `forever` so it stops after ten drops. (*Hard! You'll need a way to count, which we learn in two weeks.*)
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IF SOMETHING BREAKS

- **Flickers at the top and nothing else?** No `wait` inside the falling loop.
- **Falls once then stops?** The whole thing needs to be inside a `forever`.

- **Falls upward?** `change y by 18` should be `change y by -18`. Up is positive.
- **Scratch froze?** A `forever` with no `wait` in it. Reload and add one.